

CANADA BUILDS IT RIGHT

A new development in Okotoks, Alberta, Canada, uses seasonal solar storage to heat its homes year-round.

BY ELKA KARL

While Canada may not be the most obvious location for a solar-powered housing subdivision, the implementation of a seasonal solar storage system is making this idea a reality. Drake Landing, a new solar community that has just been built in the town of Okotoks, Alberta, outside of Calgary, is the first housing development in North America to rely on a borehole solar storage system to supply over 90% of its heating needs year-round.

The solar community, funded through a number of government and municipal grants, contains 52 two-story residences built to R-2000 and Built Green Alberta standards. The development was overseen by United Communities, which invests in building affordable homes that take advantage of emerging green technologies. Tyler Stevenson, project manager at United Communities, says that Drake Landing is an innovative, commonsense approach to affordable, healthy homes. “At United we are very aware of affordability,” says Stevenson. “Although the Calgary area is still very affordable to first-time home buyers, in recent years this affordability has eroded. We saw this as an opportunity to bring to the marketplace homes where increasing utility costs would not be an issue.”

The Homes

At this point, 52 homes have been built and sold. The public interest in the homes is very high, reports Keith Paget, the Sterling Homes representative on



The homes at Drake Landing have been built to R-2000 and Built Green Alberta standards.

the project. Sterling Homes, which is launching its own Built Green line of homes this year, constructed all of the homes at Drake Landing. “We have a wide cross-section of people [who bought the homes] but the common denominator with all of them is that they’re energy-conscious people and early adopters of technology,” says Paget. “They came to the project because they were either intrigued by the solar component or intrigued because the houses were up to R-2000 standards.”

R-2000 standards—which have been adopted worldwide—ensured the energy efficiency of the homes. “Why

would you want to build a solar home that was wasting energy?” asks Paget. “When your house uses less energy, you can also downsize your solar system slightly.” He adds that there were more practical benefits to using the R-2000 standards. “The consumer also gets a house that [looks and operates like something] they’re very familiar with. We didn’t think that the consumer was ready for some crazy-looking houses with technology that the consumer didn’t understand.” Even the unusual solar-heating system doesn’t require that the homeowner act at all unusually in order to stay warm. “When the home-

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Solar panels are placed on an 800-panel garage-mounted solar array.



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Drake Landing houses are efficient, healthy, comfortable, and operate like any standard home.

owner is in the house, they just have to set their thermostat like they usually do, and that's it. Everything's automated," says Paget.

At the time that the project was determining the house plans for Drake Landing, the Alberta Built Green program was just beginning. The program was modeled after the Colorado Built Green program, and its standards ensured that the Drake Landing homes, which are built to the Built Green Gold standards, would qualify the development as a green community. The Built Green

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Alberta standard focuses on a home's energy efficiency, indoor environmental quality (IEQ), resource use, and overall environmental impact. Canada's R-2000 standard for energy efficiency and Built Green Alberta standards ensure that the Drake Landing homes will be comfortable, energy efficient, healthy, and environmentally sensitive. Every R-2000 home is designed as a system, with structural and finishing materials, windows, doors, heating equipment, and ventilation designed to work together. They combine tight, well-insulated construction with energy-efficient heat recovery ventilation (HRV) systems. By reducing drafts and ventilating the whole house, this design significantly reduces dust, moisture, and mold. Low-emission building materials and finishes further ensure IEQ. The R-2000 standards are based on an energy consumption target

and a series of technical requirements for ventilation, airtightness, insulation, choice of materials, water use, and other factors. Each home is computer modeled and site tested by a qualified third-party R-2000 plan evaluator to determine certification.

Seasonal Thermal Storage

Despite the chilly temperatures in Canada, solar-heating systems work well there because the country receives a significant amount of solar radiation. Okotoks is a top-ten Canadian location in terms of sunny days each year—the area receives almost as much solar energy as Italy and Greece. This makes Okotoks an ideal location for solar-energy collection. However, due to Canada's geographic location and climatic conditions, the solar radiation is lower during the winter season, which is when energy use peaks. This means that the Drake Landing community requires a long-term solar storage unit in order to take advantage of the solar energy they collect.

The solar space and water heating at Drake Landing is captured year-round by an 800-panel garage-mounted solar array. On an average summer day, these panels will generate 1.5 MW of thermal power and will supply heat to the district heating system. The solar panels absorb the solar energy and heat a glycol solution running through an insulated piping system, or collector loop, that connects the array of collectors. The heated glycol is then pumped to a heat exchanger in the community's Energy Centre. The heat exchanger transfers heat to the water stored in a short-term storage tank (which holds 30,000 gallons). The glycol solution continues on through its loop back to the solar collector system.

Solar energy is collected and stored through a combination of seasonal and short-term thermal storage (STTS). The borehole thermal energy storage (BTES), an in-ground heat sink, provides the long-term seasonal energy storage, while STTS tanks provide short-term solar heating for homes. During the warmer months, the heated

water is distributed from the STTS storage tanks to the BTES system through a series of pipes made of PEX (cross-linked polyethylene), which run through a collection of 144 holes. These holes are over 115 ft (35 meters) deep and cover an area of approximately 115 ft in diameter (see Figures 1, 2, and 3).

"The borehole thermal storage system is the unique part of the project," says Paget. "There's no other long-term solar storage in North America. In Europe they've been able to achieve 60% efficiency, but they haven't had anything higher. We've taken their technology and have tried to enhance it. It's really an international collaboration."

As the heated water travels through the pipes, heat is transferred to the surrounding earth. The temperature of the earth will reach 176°F by the end of each summer. The system is designed for approximately 50% heat recovery from the BTES field. The remainder is lost to the surrounding earth because it is not possible to insulate to the depth of the borehole field.

Tests were done to confirm that ground water is not an issue. To trap the heat, the BTES is covered with sand, high-density R-40 insulation, a waterproof membrane, clay, and other landscaping materials. The water completes its circuit of the borehole system and returns to the short-term storage tanks in the Energy Centre to be heated again and repeat the same process.

"As the project group, we're very comfortable that we can meet [our heating goals]," says Paget. "The engineers, when coming up with that 90% number, felt very confident that they could attain a better number than that, and that would be on the low end of the scale. We put on 90% and try to meet or exceed the numbers that are stated. It'll take about three years to attain that 90%." The efficiency will take a few years to achieve because it will take that long for the ground mass to reach steady state, until the ground can be saturated with heat. During the time that the ground is being brought up to the final operating temperatures, losses are larger because the surrounding soil is colder than it will be under final operating conditions.

During the winter, the heated water in the BTES passes to the short-term storage tank in the Energy Centre and then circulates through the homes through the district heating loop. The district loop will vary with the outside air. The colder it gets, the higher the delivered water temperature. In the summer the delivered water would be approximately 95°F. In the winter, it would range from 95°F–131°F, and can reach 140°F if required. The design air temperature is approximately 40°F less than the water temperature due to the fan coil loss. Special fan coils have been extensively tested to accommodate the required BTU with the low temperature heat. The heated water in each home passes through a heat exchanger within a specially designed, low-temperature air handler unit located in the basement. A fan, also within the unit, blows air across the warm coil. Heat is passed from water to air and then distributed throughout the house via the home's ductwork. When the temperature of the home's thermostat is met, an



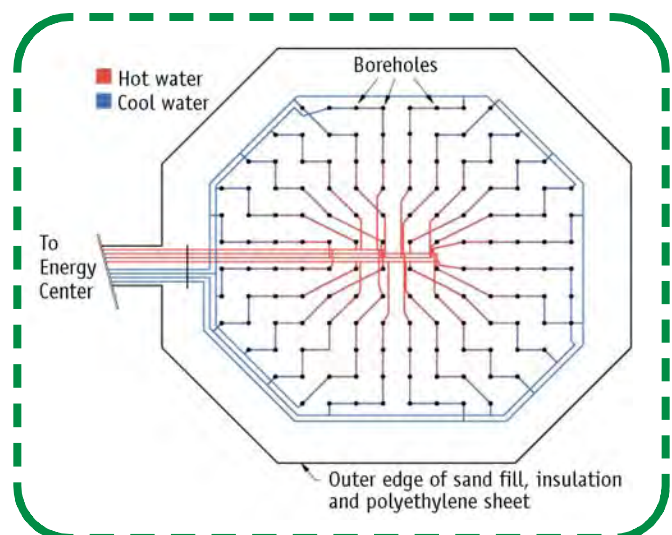
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The borehole thermal energy storage system is made up of a series of pipes that run through 144 holes sunk 120 ft deep.



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Although these homes are using an innovative solar technology, homeowners need only adjust their thermostat to use the technology--no special training is required.



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Figure 1. Plastic, insulated, underground pipe is used to distribute heated water from the community's Energy Centre back to the homes.

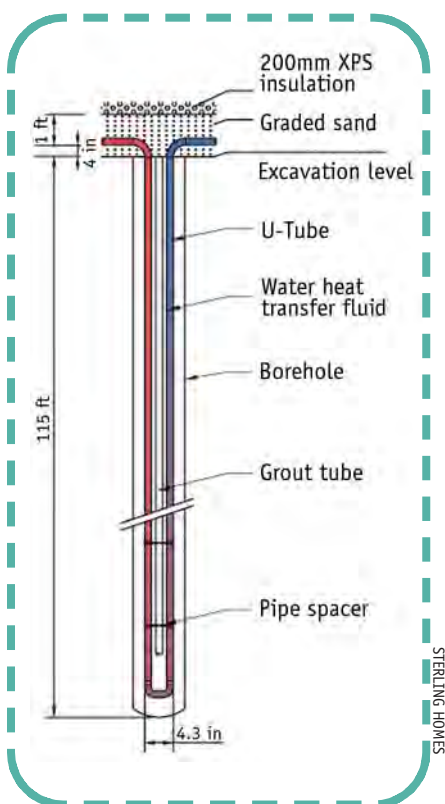


Figure 2. The Borehole Thermal Energy Storage (BTES) consists of 144 boreholes.

automatic valve in the basement shuts off the heat transfer between the district heating loop and the air handler unit. If the STTS still cannot meet the demand for heat, the Energy Centre's backup gas boiler will switch on to increase the temperature.

Good for Homeowner and Community

As a nation participating in the Kyoto Protocol, Canada benefits from communities such as Drake Landing because they cut greenhouse gas emissions through their innovative energy efficiency standards and solar heating. In fact, the most immediate benefit of Drake Landing will be a decrease in greenhouse gas emissions of approximately 5 tons per home per year. This reduction goes above and beyond the Canadian government's 1-Ton Challenge goal and, over the life of the system, should reduce greenhouse gas emissions by about 13,000 tons while providing comfortable homes with excellent IEQ for 52 families.

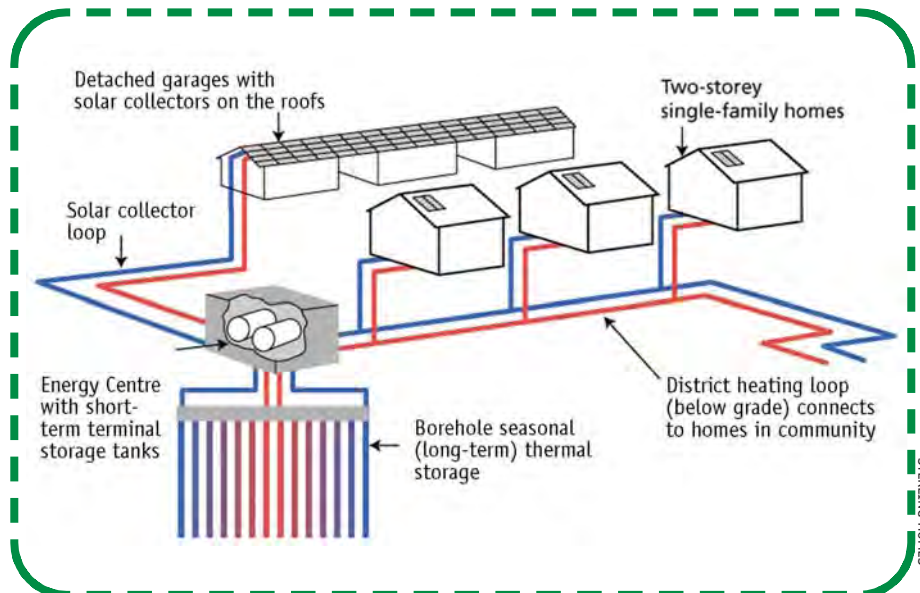


Figure 3. After three years, 90% of the homes' heating needs should be met by the BTES system.

Although the system's costs may not be competitive with today's price for fossil fuels, such as natural gas, the scale of the demonstration will be large enough to make it competitive with higher-priced conventional heating sources such as electricity. In addition, as these projects grow larger and fossil fuel prices increase, the costs of the systems will become more competitive. The operating costs will be lower than those of a combustion furnace. In addition, the Drake Landing project will provide invaluable information for future, larger-scale projects.

United Communities is currently conducting a study to determine the feasibility of bringing a communitywide geothermal heating system to future homes in Drake Landing. Says Stevenson, "The town of Okotoks has been very progressive in their views around energy efficiency, hence providing the perfect environment to conduct such a test. With items like solar heating for the town's local pool and recycling depot, water conservation measures, and now the Drake Landing solar community, a community of geothermal-heated homes seems like the next logical step. And with natural gas prices' continuous climb, the market has been very receptive to our energy-efficient initiatives."



*Elka Karl is an associate editor at **Home Energy** magazine.*

Canadian organizations providing expertise and funding for the project include Natural Resources Canada (NRCan), whose Technology Centre initiated and leads this project; Technology Early Action Measures (TEAM), a government program designed to develop technologies that mitigate greenhouse gas emissions and help sustain economic and social development; Environment Canada's Meteorological Services, Atlantic Region, Canada's expert in underground thermal storage technologies; the Federation of Canadian Municipalities; and the government of Canada.

FOR MORE INFORMATION:

To learn more about Built Green Alberta, go to www.builtgreenalberta.com.

To learn more about R-2000 standards, go to <http://r2000.chba.ca>.

To learn more about Drake Landing, go to www.dlsc.ca.